Statement of Intent

UCD School of Physics

Advanced Laboratories

Name: Dearbhail O'Mahony
Experiment: Muon Detector
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The aim of this lab is to detect cosmic ray muons and to use the detector setup to generate a muon decay curve and estimate muon lifetime. There will be a possible extension to characterise the muon flux dependence on zenith angle.

To form the detector, a scintillator will be connected to a photomultiplier tube, and in turn a high voltage source. Pulses from the scintillator will go through an amplifier and then a discriminator. When a muon is trapped in the scintillator, this pulse will trigger the discriminator and send a pulse into a time-amplitude converter (TAC) which will cause a 'start' signal. When a muon stopped in the scintillator decays, the resulting electron will cause another pulse to be sent to the TAC, corresponding to a 'stop' signal. The TAC will then output a pulse with an amplitude linearly proportional to the start-stop time, which will go to a multi-channel analyser, where the data will be sorted into channels corresponding to this time. Plotting this data (counts in each channel against channel number, i.e. time between events) will give a decay curve, and by using a log scale for the y-axis the muon lifetime can be ascertained from the slope of this plot by $lifetime = -1/slope$.

For the extension, measurements will be made using a cosmic ray telescope consisting of two detectors and photomultiplier tubes. Tests will be made with this equipment and if time and the available equipment allows, the count rate of the telescope, averaged over an hour, will be measured for a range of angles between 0 ° and 90°, and plotted to demonstrate the effect of zenith angle on muon flux. This can be compared to the theoretical prediction given by the equation $I(\theta) = I_0 \cos^n \theta$, where I_0 is the intensity at a zenith angle of 0°.

The safety considerations for this lab are to ensure that all equipment is wired up correctly so that no damage occurs, and for care to be taken with the high voltage supply, ensuring that it is switched off when not required or when the setup is being altered.